

Intelligent Agents; Cybernetic Intelligence vs Artificial Intelligence

Artificial Intelligence

Agent

An *agent* is anything that can be viewed as perceiving its environment through *sensors* and acting upon that environment through *actuators*.

This definition is deliberately broad it covers

A human (eyes, ears as *sensors*; hands, legs as *actuators*),

A robot (cameras, infrared *sensors*; motors, grippers),

A software agent (keystrokes or file contents as percepts; screen output or network packets as actions) equally well.

Agent

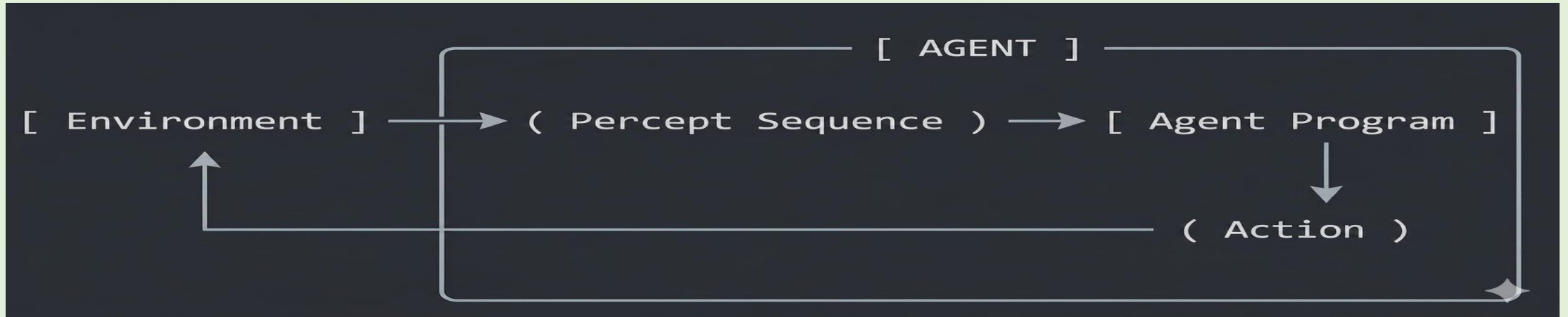
The sequence of everything an agent has perceived so far is called its *percept sequence*.

An *agent's* behavior is mathematically described by an *agent function*,
The mathematical description of the agent's behavior. It is an abstract mapping that takes any given percept sequence and determines the corresponding action to take:

$$f: P^* \rightarrow A,$$

where *f* is the agent function, *P*^{*} represents the complete history of perceptions, and *A* is the resulting action.

Agent



In practice this function is implemented by an *agent program*, which is the actual code running on some physical architecture.

The *agent* function is an abstract, often infinite, mathematical mapping; the *agent* program is the concrete, finite implementation of it.

Rational Agents

A *rational agent* is one that, for each possible percept sequence, selects an action expected to maximize its *performance measure*, given the evidence provided by the percept sequence so far and whatever built-in knowledge the agent has.

Four factors determine rationality at any given moment:

The performance measure that defines the criterion of success.

Everything the agent has perceived so far (its complete percept sequence).

What the agent knows about the environment, prior to and independent of experience.

The actions the agent is capable of performing.

PEAS

Before designing any *agent*, its task environment needs to be specified precisely, and Russell and Norvig group this specification into four components, remembered by the acronym *PEAS*:

Component	Meaning
Performance measure	The criteria that define success for the agent
Environment	The world the agent operates within
Actuators	The mechanisms through which the agent acts
Sensors	The mechanisms through which the agent perceives

PEAS

Consider an automated taxi driver:

PEAS Component	Automated Taxi Example
Performance measure	Safety, speed, legality, comfortable trip, profit maximization
Environment	Roads, other traffic, pedestrians, customers
Actuators	Steering, accelerator, brake, signal, horn
Sensors	Cameras, sonar, speedometer, GPS, engine sensors, keyboard/microphone for passenger input

Properties of Task Environments

Environments themselves can be classified along several independent dimensions, and where an environment falls on each dimension has a direct effect on how hard the agent-design problem is and which techniques from later in this course actually apply.

Dimension	Description	Example (Simple)	Example (Complex)
Fully vs. Partially Observable	Can sensors access the complete state of the environment at every point?	Chess (fully observable board)	Poker (opponents' hands hidden)
Single vs. Multi-agent	Is the agent operating alone or alongside other agents?	Crossword puzzle solving	Chess, driving in traffic
Deterministic vs. Stochastic	Is the next state fully determined by current state and action, or is there uncertainty?	Chess (deterministic)	Backgammon (dice involved)
Episodic vs. Sequential	Is each action independent of past actions, or do decisions have long-term consequences?	Defect-detection on an assembly line (episodic)	Chess, driving (sequential)
Static vs. Dynamic	Can the environment change while the agent is deliberating?	Crossword puzzle (static)	Driving a taxi (dynamic)
Discrete vs. Continuous	Are states, time, and percepts/actions distinct and countable, or continuously varying?	Chess (discrete positions)	Taxi driving (continuous speed, position)
Known vs. Unknown	Does the agent (not the environment itself) already know the rules/outcomes of its actions?	Solitaire with known rules	Learning to play a brand-new video game